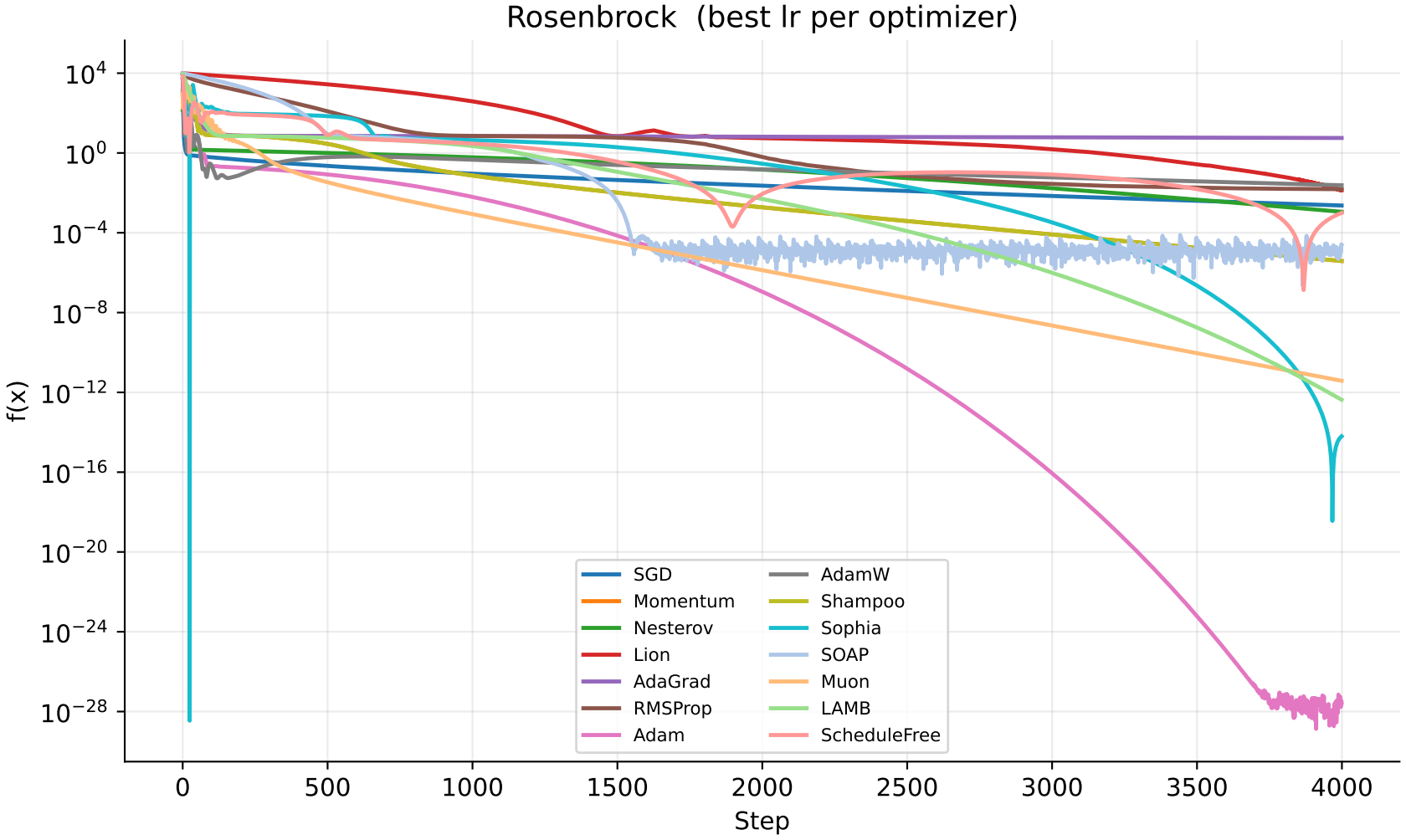
Rosenbrock (best lr per optimizer)
f(x)
Step
10^{4}
10^{0}
10^{-4}
10^{-8}
10^{-12}
10^{-16}
10^{-20}
10^{-24}
10^{-28}
0
500
1000
1500
2000
2500
3000
3500
4000
SGD
Momentum
Nesterov
Lion
AdaGrad
RMSProp
Adam
AdamW
Shampoo
Sophia
SOAP
Muon
LAMB
ScheduleFree